

LC 1249: Min Remove to Make Valid Parentheses

© Authored by: Arpit Raj, LNMIIT Jaipur

Date: September 11, 2026 | Tags: GE digital, Tencent

Objective

This time need to return actual string after doing the removal operation.

Intuition

```
) ( ( → " "
```

because both `)` and `(` were invalid, hence need a stack to push invalid indices, stack because whether valid or not depends on future checks.

i) `(` → push it's index for removal

`)` → needs a previous `(` so if stack isn't empty, match it and pop, `)` saved `(`

If stack empty, then `)` was invalid. mark it's index for removal, dont add to stack though as it reserved for tracking invalid `(`.

Hence:

- one stack for invalid `(`
- one stack for invalid `)`
- one vector to mark all invalid indices

C++ Solution

```

class Solution {
public:
    string minRemoveToMakeValid(string s) {
        int n = s.size();
        stack<int> st1;
        stack<int> st2;
        vector<int> remove(n, 0);

        for(int i = 0; i < n; i++) {
            char c = s[i];

            if(c == '(') {
                st1.push(i);
            }
            else if(c == ')') {
                if(!st1.empty()) {
                    st1.pop();
                }
                else {
                    st2.push(i);
                }
            }
        }

        while(!st1.empty()) {
            remove[st1.top()] = 1;
            st1.pop();
        }

        while(!st2.empty()) {
            remove[st2.top()] = 1;
            st2.pop();
        }

        string ans = "";
        for(int i = 0; i < n; i++) {
            if(remove[i] == 0) {
                ans += s[i];
            }
        }

        return ans;
    }
};

```